

ZEPTO DATA ANALYSIS PROJECT

SQL Data Exploration, Cleaning & Business Intelligence Dashboard

PROJECT REPORT

Project Dataset	Zepto Product Dataset
Primary Tools	PostgreSQL / SQL, Power BI
Analysis Type	Exploratory Data Analysis & Business Insights
Prepared From	Provided CSV dataset, SQL analysis script and Power BI dashboard

1. Executive Summary

This project analyzes a Zepto product dataset to understand product categories, pricing, discounts, stock availability, inventory quantity and estimated revenue. The work combines SQL-based data exploration and cleaning with a Power BI dashboard for visual business analysis.

The SQL workflow begins with understanding the dataset structure and quality, then checks missing values, categories, stock status and duplicate product names. Data preparation includes removing records with zero MRP and converting price values from paise to rupees. Business queries are then used to identify highly discounted products, expensive out-of-stock products, category-level estimated revenue, average discounts, price per gram and inventory weight.

2. Dataset Overview

Rows in CSV	3,732
Columns	9
Product identifier	sku_id
Main product fields	category, name, mrp, discountPercent, discountedSellingPrice
Inventory fields	weightInGms, availableQuantity, outOfStock, quantity

The SQL table is named zepto and defines sku_id as the primary key. The project script uses numeric fields for prices, discounts, quantities and product weight, together with a Boolean stock-status field. [Filecite](#) [turn0file0](#) [L1-L10](#)

3. Project Objectives

- Understand the structure and quality of the Zepto product dataset.
- Identify missing, zero-price and repeated product records.
- Clean price values and prepare the data for analysis.
- Analyze discounts, pricing and product value.
- Examine stock availability and high-value out-of-stock products.
- Estimate category-level revenue using selling price and available quantity.
- Analyze inventory weight and classify products by weight.
- Present the analysis in an easy-to-understand Power BI dashboard.

4. Methodology

1. Data Loading	The product dataset is loaded into a structured SQL table.
2. Data Exploration	Row count, sample records, categories, stock status and repeated product names are examined.
3. Data Quality Check	The SQL script checks important columns for NULL values and identifies products with zero prices.
4. Data Cleaning	Rows with MRP equal to zero are removed and price values are converted from paise to rupees.
5. SQL Analysis	Business questions are answered using filtering, grouping, aggregation, sorting, CASE expressions and calculated fields.
6. Dashboarding	The prepared data and analysis are used for Power BI-based visualization and reporting.

5. SQL Analysis Performed

5.1 Data Exploration

The project checks the number of records, views sample data, detects NULL values, lists distinct product categories, compares in-stock and out-of-stock products, and identifies product names associated with multiple SKUs.

These checks establish the basic structure and quality of the dataset before business analysis.

[filecite\turn0file0\L14-L58](#)

5.2 Data Cleaning

Products with MRP equal to zero are removed. The script then divides MRP and discounted selling price by 100 to convert values from paise to rupees. This makes the pricing values easier to interpret in the report.

The cleaning logic is explicitly implemented in the provided SQL script. [filecite\turn0file0\L60-L74](#)

5.3 Business Analysis Queries

Top 10 discounted products	Uses discount percentage to identify products offering the largest discounts.
High-MRP out-of-stock products	Finds out-of-stock products with MRP above ₹300 and sorts them by MRP.
Estimated revenue by category	Calculates category-level estimated revenue as discounted selling price × available quantity.
High-MRP, low-discount products	Finds products with MRP above ₹500 and discount below 10%.
Highest average category discounts	Returns the five categories with the highest average discount percentage.
Price per gram	Calculates discounted selling price per gram for products weighing at least 100 g.
Weight classification	Groups products into Low, Medium and Bulk using weight thresholds.
Total inventory weight	Calculates total inventory weight by multiplying product weight by available quantity.

The above analyses correspond directly to the SQL queries supplied with the project. [filecite\turn0file0\L76-L132](#)

6. Power BI Dashboard

The project also includes a Power BI dashboard file. The dashboard is intended to transform the prepared product data into visual summaries that can be used to understand product pricing, discounts, inventory and category-level performance.

- Category-level analysis
- Pricing and discount analysis
- Stock and availability analysis
- Inventory-oriented metrics
- Business-focused visual summaries

The supplied PBIX file is the dashboard source. This report does not invent visual names, KPI values or chart results that are not directly available in the supplied project materials.

7. Key Analytical Insights

- Discount percentage can be used to identify products with stronger promotional pricing.
- Out-of-stock products with relatively high MRP can highlight potentially important availability gaps.
- Estimated revenue can be compared across categories using discounted selling price and available quantity.
- Price-per-gram analysis provides a normalized way to compare product value among products of different weights.
- Weight classification makes inventory analysis easier by separating products into Low, Medium and Bulk groups.
- Total inventory weight provides a category-level view of the physical quantity represented by available stock.

These observations describe what the SQL metrics are designed to reveal; they are not claims of operational sales performance.

8. Technical Implementation

Database	PostgreSQL / SQL
Visualization	Microsoft Power BI
Dataset format	CSV
Core SQL techniques	SELECT, WHERE, GROUP BY, HAVING, ORDER BY, DISTINCT, aggregate functions, CASE
Cleaning operations	DELETE and UPDATE
Calculated metrics	Estimated revenue, average discount, price per gram, total inventory weight

The SQL script demonstrates the use of aggregation, filtering, conditional classification and calculated measures.

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9. Conclusion

The Zepto Data Analysis project provides a compact end-to-end example of turning raw product data into useful business-oriented information. SQL is used to inspect and clean the data and to answer targeted analytical questions, while Power BI provides a visual layer for communicating the results.

The project can support further analysis such as category performance tracking, discount strategy analysis, inventory monitoring and product-level value comparison when additional business data becomes available.

10. Future Scope

- Add time-based sales and inventory data for trend analysis.
- Track stock changes over time and identify frequently unavailable products.
- Add customer/order data to calculate actual revenue instead of estimated inventory-based revenue.
- Create interactive Power BI filters for category, stock status, price range and discount range.
- Develop automated refresh pipelines for regularly updated product data.

11. Appendix: Main SQL Operations

The provided SQL script contains operations for table creation, exploration, cleaning and business analysis. Selected operations include:

```
SELECT COUNT(*) FROM zepto;
```

```
SELECT DISTINCT category FROM zepto ORDER BY category;
```

```
SELECT outOfStock, COUNT(sku_id) FROM zepto GROUP BY outOfStock;
```

```
UPDATE zepto SET mrp = mrp / 100.0, discountedSellingPrice = discountedSellingPrice / 100.0;
```

```
SELECT category, SUM(discountedSellingPrice * availableQuantity) AS total_revenue FROM zepto GROUP BY category;
```

```
ROUND(discountedSellingPrice / weightInGms, 2) AS price_per_gram
```